Work Plan: October 30 to December 31, 2026

Smart, Life-Aware IoT Health Monitoring: An AI-Optimized Communication Architecture for Local Body-Area Patient Monitoring

1. Scope and Assumptions

- Architecture follows the project proposal: five layers (sensing, understanding, acting, recording, viewing).
- One AI model (patient state: Normal / At-Risk / Emergency) plus a rule-based communication controller that sets transmission rate and channel.
- Upgrade if time allows: train a Decision Tree on graded scenarios to replace the rule table.
- Sensor node and gateway talk over ESP-NOW. Gateway talks to the backend over Wi-Fi (HTTP).
- Before any hardware is ready, every team starts with simulated data, so no one waits for anyone else.

2. Team and Responsibilities

Sub-team	Members	Responsibility
Embedded	• Ahmed Abdelrahman Ahmed Fouad • Islam Awad Hassan Mohamed	Layer 1: sensors, ESP-NOW link, channel scan and switching, applying controller decisions
AI and Database	• Ahmed Ayman Fathy Mahrous	Layer 2: patient state model (Model 1). Layer 4: database and evaluation numbers
Backend	• Mohamed El-Sayed Antar Hussein • Abdullah Aziz Fathy	Layer 4: FastAPI endpoints, logging, metrics
Front-end	• Mohamed Tarek Mohamed Hassan • Mohamed Ahmed Hassan Mohamed • Ahmed Yasser Ahmed Fouad	Layer 5: live dashboard for the care team

3. Milestones

Milestone	Target date	What must work
Milestone 1	Nov 29	Sensor node to gateway to backend to dashboard, with Model 1 state shown and the transmission rate changing with patient state
Milestone 2	Dec 20	Channel selection working, evaluation runs done, model fine-tuned, wearable assembled
Final delivery	Dec 31	Code frozen, report and slides ready, backup demo video recorded

4. Weekly Plan

Week	Dates / Focus	Embedded (Layer 1)	AI and Database (Layer 2 and 4)	Backend (Layer 4)	Front-end (Layer 5)
Week 1	**Oct 30 – Nov 8** *Setup*	• Receive hardware order • Buy AD8232 and a 10k potentiometer • Test each sensor on its own	• Download Kaggle / PhysioNet datasets • Clean data and explore features	• FastAPI skeleton • SQLite table "readings" • Endpoint POST /api/reading	• Dashboard layout with fake data • Team: shared Git repo and agreed JSON packet format
Week 2	**Nov 9 – Nov 15** *Layers in isolation*	• All sensors read on one node • ESP-NOW node to gateway with packet_id	• Train Model 1 v1 (Random Forest) • Report accuracy per class	• GET /api/history • GET /api/metrics (first version)	• Live charts for BPM and SpO2 • Color-coded status banner
Week 3	**Nov 16 – Nov 22** *First integration*	• Gateway sends readings to backend over HTTP • Packet-loss calculation from packet_id	• Export Model 1 with joblib • Help integrate it into the backend	• Run Model 1 inside the backend • Store patient state with each reading	• Dashboard fed by polling every 2 s • Show patient state from backend
Week 4	**Nov 23 – Nov 29** *Controller*	• Measure RSSI and packet loss on gateway • Rule table sets transmission rate • Node applies the new rate	• Support controller thresholds • Log decisions to database	• Store controller decisions • Metrics endpoint: PDR and latency	• Show RSSI, channel and rate on dashboard • MILESTONE 1: end-to-end demo to the doctor
Week 5	**Nov 30 – Dec 6** *Channel selection*	• Scan, score and select best channel (1 / 6 / 11) • Switch both ends to the same channel	• Collect real data from our own hardware • Label it for fine-tuning	• Evaluation_runs table • Store collected data	• Event log for state changes • Optional: WebSocket upgrade
Week 6	**Dec 7 – Dec 13** *Evaluation*	• Run tests: interference, distances, simulated emergency • Both baseline and optimized runs	• Compute PDR, latency, throughput, energy estimate • Prepare comparison charts	• Metrics endpoint finalized • Bug fixes from test runs	• Dashboard polish • Page for with-AI vs without-AI comparison
Week 7	**Dec 14 – Dec 20** *Refinement*	• Wearable assembly on battery • Power and stability checks	• Fine-tune Model 1 on own data • Final accuracy numbers	• Stability and error handling • Prepare backup demo data	• Final UI fixes • MILESTONE 2: near-final version
Week 8	**Dec 21 – Dec 27** *Documentation*	• Hardware section of report • Wiring diagram	• Models and evaluation section • Limitations (public vs own data, BER)	• Architecture and API section • Database schema	• Screenshots and slides • Rehearse the demo several times

Week	Dates / Focus	Embedded (Layer 1)	AI and Database (Layer 2 and 4)	Backend (Layer 4)	Front-end (Layer 5)
Week 9	**Dec 28 – Dec 31** *Buffer*	• Fix any last hardware issue	• Final model files frozen	• Code freeze • Backup copy of everything	• Record a backup demo video

5. Evaluation Plan (Weeks 6 and 7)

- Run the same physical test twice: baseline (fixed rate, fixed channel) and optimized (Model 1 plus controller).
- Conditions: induced interference, varying distances between node and gateway, simulated emergency readings.
- Metrics: Packet Delivery Ratio (target 95% or more in emergencies), latency, energy consumption estimate, throughput.
- Success criterion: higher PDR and throughput and lower latency in emergencies, and lower average energy in normal states.

6. Risks and Mitigation

Risk	Mitigation
Hardware delivery delay	Develop backend, model and dashboard with simulated data first; hardware joins when it arrives
Public dataset differs from our sensors	Train on public data, then fine-tune on data from our own hardware; state this limitation in the report
Bit Error Rate is hard to measure on ESP-NOW	Rely on PDR and packet loss as the main metrics; document BER as a known limitation
WebSocket takes too long	Keep polling every 2 seconds; the demo result is the same
Exams in December or January	Dates for Weeks 7 and 8 can shift; Week 9 is kept as a buffer